

**AMENDMENT NO. 1   APRIL 2002**  
**TO**  
**IS 1538 :1993   CAST IRON FITTINGS FOR PRESSURE**  
**PIPES FOR WATER, GAS AND SEWAGE —**  
**SPECIFICATION**  
*( Third Revision )*

( Page 1, clause 4.3 ) — Substitute ‘230 HBS’ for ‘210 HBS’.

( Page 1, clause 4.3 ) — Insert the following new clause after 4.3:

‘ 4.3.1 In case hardness is more than 230 HBS, destructive test shall be carried out for observing fracture which shall be grey without any chilling effect.’

( Page 2, Table 1 ) — Substitute ‘kgf/cm<sup>2</sup>’ for ‘N/mm<sup>2</sup>’ at both the places.

( Page 2, clause 10.4.1 ) — Insert the following new clauses after 10.4.1:

**‘10.5 Tolerance on Height of Raised Face (c) in mm**

| <i>Height of Raised<br/>Face (c)</i> | <i>Tolerance</i> |
|--------------------------------------|------------------|
| 3                                    | +1.5<br>-2.0     |
| 4                                    | +2<br>-3.0       |
| 5                                    | +2.5<br>-4.0     |

**Amend No. 1 to IS 1538 : 1993**

**10.6 Tolerances on the External Diameter 'D' in mm**

|                            |                              |                        |                            |
|----------------------------|------------------------------|------------------------|----------------------------|
| <i>DN<br/>Tol<br/>on D</i> | 80, 100, 125                 | 150, 200, 250,<br>300  | 350, 400, 450,<br>500, 600 |
|                            | $\pm 4.5$                    | $+5.5$<br>$-2.5$       | $+6.5$<br>$-3$             |
| <i>DN<br/>Tol<br/>on D</i> | 700, 750, 800,<br>900, 1 000 | 1 050, 1 100,<br>1 200 | 1 500                      |
|                            | $+7.5$<br>$-4$               | $+8.5$<br>$-4$         | $+10$<br>$-5$              |

**10.7 Tolerance on Flange Drillings in mm**

| <i>Dimension</i>                           | <i><u>Bolt Hole Diameter</u></i> |              |
|--------------------------------------------|----------------------------------|--------------|
|                                            | 19 to 28                         | 31 to 43     |
| Bolt hole diameter, <i>d</i>               | $+2$<br>$-0$                     | $+3$<br>$-0$ |
| Pitch circle diameter, <i>C</i>            | $\pm 2$                          | $\pm 2.8$    |
| Centre to centre of<br>adjacent bolt holes | $\pm 2$                          | $\pm 2.8$    |

NOTE — Tolerance on all other untoleranced dimensions shall be as per IS 5519 : 1979  
'Deviations for untoleranced dimensions and mass of grey iron castings'.

( Page 3, clause 12.5, line 3 ) — Insert 'for 5 minutes' after '65°C'.

( Page 3, clause 12.5 ) — Insert the following new clause after 12.5:

**'12.5.1** Coating test shall be conducted on a sample piece cut from the pipe having a sample area not less than 10 sq.cm.'

( Page 3, clause 14.1 ) — Substitute the following for the existing:

**'14.1** The dimensions and designs for fittings suitable for lead joint and flanged joints covered in this standard shall conform to Tables 3 to 6. The dimensions of socket and spigot of pipes in lead joints given in Table 2 is for reference only.'

( MTD 6 )